CSE472:DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: Understand the origin, evolution, and linguistic foundations of Natural Language Processing

CO2 :: Understand word embeddings, semantic relationships, and embedding visualization techniques

CO3 :: Apply deep learning sequence models for NLP to text and sentiment classification tasks

CO4 :: Apply seq2seq models with attention for translation and summarization tasks

CO5 :: Use transformer architectures and pretrained language models for fine-tuned NLP tasks

CO6 :: Apply generative NLP and large language models for text generation and evaluation tasks, considering explainability and hallucination

Unit I

Foundations of NLP : origin of NLP, language and grammar, linguistic essentials, morphology, syntax, semantics, challenges of NLP, applications of NLP

Text Processing : tokenization, stemming, lemmatization, stop-word removal, punctuation handling, handling out-of-vocabulary words, normalization, Bag-of-Words, n-grams, TF-IDF

Unit II

Word Embeddings and Vector Representations : vector space models, dense word embeddings, Word2Vec, CBOW, Skip-Gram models, GloVe embeddings, capturing semantic similarity, analogy relationships, visualizing embedding spaces using PCA or t-SNE

Unit III

Deep Learning Sequence Models for NLP : sequential text data, recurrent neural networks, long short term memory networks, gated recurrent units, bidirectional RNNs, sequence modeling applications, sentiment classification, text classification, sequence training techniques, teacher forcing, truncated backpropagation through time, evaluation metrics for sequence tasks

Unit IV

Sequence-to-Sequence Models and Attention Mechanisms : encoder–decoder architectures for NLP, sequence-to-sequence models for machine translation and summarization, attention in deep NLP, soft attention, Bahdanau and Luong attention, integrating attention into encoder–decoder networks, evaluation techniques, BLEU and ROUGE scores, limitations of classical seq2seq models

Unit V

Transformers : transformer architecture, self-attention, multi-head attention, positional encoding, transformer encoder and decoder blocks, tokenization methods, Byte-Pair Encoding and WordPiece

Pretrained Language Models : pretrained transformer models, BERT, GPT, T5, masked language modeling, next sentence prediction, causal language modeling, transfer learning for NLP tasks, fine-tuning for text classification, named entity recognition, question answering, HuggingFace Transformers

Unit VI

Generative NLP and LLMs : generative NLP models, text generation strategies, greedy search, beam search, top-k, nucleus sampling, instruction-tuned, large language models, model behaviors in summarization, dialogue generation, reasoning tasks, evaluation metrics, perplexity and human judgment measures, explainability and hallucination in LLMs

List of Practicals / Experiments:

List of Practicals

- Given a set of text documents, implement a preprocessing pipeline that performs tokenization, lowercasing, noise removal, stopword removal, stemming/lemmatization, and converts the processed text into numerical representations using Bag-of-Words and TF-IDF
- Load pretrained Word2Vec or GloVe embeddings, select a set of words, compute their embeddings, and visualize relationships in 2-D space using PCA or t-SNE
- Train CBOW and Skip-Gram Word2Vec models on a sample text corpus, compute word similarities and analogies, and compare the performance of both models

- Load pretrained GloVe or Word2Vec embeddings, select word groups belonging to different semantic categories, reduce dimensionality using PCA or t-SNE, and visualize the embeddings to interpret clusters and relationships
- Build a sentiment analysis model using an LSTM network on a dataset such as the IMDB movie reviews or SMS Spam dataset. Compare its performance with a baseline model that uses Bag-of-Words or TF-IDF features
- Implement a Bidirectional LSTM model for Named Entity Recognition using the CoNLL-2003 dataset or a smaller annotated dataset. Train the model to label words as Person, Location, Organization, or Other
- Implement a sequence-to-sequence model with Bahdanau or Luong attention to translate short English sentences into another language using a small parallel dataset. Compare translation quality with and without attention
- Build an attention-based sequence-to-sequence model to generate short summaries from news headlines or paragraph datasets. Train briefly using a subset of the dataset and evaluate generated summaries using ROUGE scores
- Fine-tune a pretrained BERT model for sentiment classification on a dataset such as the IMDB movie reviews or the SMS Spam dataset. Evaluate the model using accuracy and F1-score, and analyze how BERT's contextual embeddings improve classification performance
- Use a pretrained GPT-2 model to generate text based on a given prompt. Apply different multiple decoding strategies and compare the fluency, creativity, and coherence of the generated outputs

Text Books:

1. GETTING STARTED WITH DEEP LEARNING FOR NATURAL LANGUAGE PROCESSING by SUNIL PATEL, BPB Publication, New Delhi

References:

1. SPEECH AND LANGUAGE PROCESSING: AN INTRODUCTION TO NATURAL LANGUAGE PROCESSING, COMPUTATIONAL LINGUISTICS AND SPEECH RECOGNITION by DANIEL JURAFSKY, Pearson Education India

CSE357:COMBINATORIAL STUDIES

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamental computer science concepts, including data structures, algorithms, databases, operating systems, and computer networks, essential for technical interviews.

CO2 :: assess the problem-solving skills specific to coding challenges and algorithmic problems frequently encountered in technical interviews.

CO3 :: articulate a comprehensive command of Object-Oriented Programming principles, enhancing readiness to excel in technical interviews.

CO4 :: gain comprehensive strategies and techniques to effectively prepare for technical interviews, including mock interviews, resume building, and effective communication during interviews.

CO5 :: impart the requisite knowledge and skills to confidently address technical interview inquiries posed by service-oriented IT companies, demonstrating proficiency across diverse technical domains.

CO6 :: prepare and practice sessions including mock interviews, whiteboard coding exercises, and simulations of behavioural interview scenarios, fostering confidence and preparedness for different stages of technical interviews.

Unit I

Operating System Basics : Foundations of Operating Systems, Types of Operating Systems, Memory Management, Processor Scheduling Algorithms, Process Synchronization and Inter-process Communication

Unit II

Computer Networking Basics : Foundations of Computer Networks,, Network Topologies and Transmission Medium,, Understanding OSI and TCP/IP Models,, Subnetting and Routing,, Key Protocols: HTTP, SMTP, POP/IMAP, FTP, DNS, DHCP,

Unit III

Database Management Systems (DBMS) : Introduction to Databases and RDBMS,, Data Definitions: Tables, Fields, Records,, SQL and Data Manipulation,, Database Keys and Data Integrity,, Database Normalization and Transactions,, Transaction Management in DBMS

Unit IV

Fundamentals of Programming Languages : C/C++/Java Interview Questions Overview,, Pointers and Storage, Classes,, Introduction to Object-Oriented Programming (OOP),, Techniques of Parameter Passing and Binding,, Memory Handling in OOP Languages

Unit V

Data Structures : Introduction to Data Structures, Linked Lists (Types and Operations), Stacks and Queues (Operations and Applications), Trees(Binary Trees and Balanced Trees), Graphs (Representation and Algorithms), Hashing (Applications and Functions), Heaps(Applications and Operations)

Unit VI

Algorithms : Understanding Algorithms and Analysis, Running Time Analysis and Rate of Growth, Asymptotic Notation: Big-O Notation, Recursion and Backtracking, Dynamic Programming Strategies, Greedy Algorithms

List of Practicals / Experiments:

List of Practicals

- 1. Introduction to Linux Commands
- 2. Shell Programming
- 3. System Calls (File / Directory/ Process)
- 4. Multi Thread Process using Pthread Library
- 5. SQL Commands (DDL/DML/ TCL)

- 6. PL/SQL Triggers, Procedures, Functions
- 7. Subnetting
- 8. Arrays and Linked List
- 9. Stacks and Queues
- 10. Trees and Binary Search Trees

Text Books:

1. IT INTERVIEW QUESTIONS by NARASIMHA KARUMANCHI, CAREERMONK PUBLICATIONS

References:

1. MCQS IN COMPUTER SCIENCE by TIMOTHY WILLIAMS, MC GRAW HILL

2. SQL IN 10 MINUTES, SAMS TEACH YOURSELF by BEN FORTA, SAMS PUBLISHING

3. CRACKING THE IT INTERVIEW by M BALASUBRAMANIAM, K. R BAALAJI, KIRAN. G RANGANATH, NANDAWAT RAVINDRA K, M SELVAGURU, T COMERICA SUBASH, RAGHAVAN S VENKAT, S ANBAZHAGAN VIKRAM, MC GRAW HILL

4. CRACKING THE CODING INTERVIEW by MCDOWELL GAYLE LAAKMANN, CAREERCUP

CSE357:COMBINATORIAL STUDIES

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: understand the fundamental computer science concepts, including data structures, algorithms, databases, operating systems, and computer networks, essential for technical interviews.

CO2 :: assess the problem-solving skills specific to coding challenges and algorithmic problems frequently encountered in technical interviews.

CO3 :: articulate a comprehensive command of Object-Oriented Programming principles, enhancing readiness to excel in technical interviews.

CO4 :: gain comprehensive strategies and techniques to effectively prepare for technical interviews, including mock interviews, resume building, and effective communication during interviews.

CO5 :: impart the requisite knowledge and skills to confidently address technical interview inquiries posed by service-oriented IT companies, demonstrating proficiency across diverse technical domains.

CO6 :: prepare and practice sessions including mock interviews, whiteboard coding exercises, and simulations of behavioural interview scenarios, fostering confidence and preparedness for different stages of technical interviews.

Unit I

Operating System Basics : Foundations of Operating Systems, Types of Operating Systems, Memory Management, Processor Scheduling Algorithms, Process Synchronization and Inter-process Communication

Unit II

Computer Networking Basics : Foundations of Computer Networks,, Network Topologies and Transmission Medium,, Understanding OSI and TCP/IP Models,, Subnetting and Routing,, Key Protocols: HTTP, SMTP, POP/IMAP, FTP, DNS, DHCP,

Unit III

Database Management Systems (DBMS) : Introduction to Databases and RDBMS,, Data Definitions: Tables, Fields, Records,, SQL and Data Manipulation,, Database Keys and Data Integrity,, Database Normalization and Transactions,, Transaction Management in DBMS

Unit IV

Fundamentals of Programming Languages : C/C++/Java Interview Questions Overview,, Pointers and Storage, Classes,, Introduction to Object-Oriented Programming (OOP),, Techniques of Parameter Passing and Binding,, Memory Handling in OOP Languages

Unit V

Data Structures : Introduction to Data Structures, Linked Lists (Types and Operations), Stacks and Queues (Operations and Applications), Trees(Binary Trees and Balanced Trees), Graphs (Representation and Algorithms), Hashing (Applications and Functions), Heaps(Applications and Operations)

Unit VI

Algorithms : Understanding Algorithms and Analysis, Running Time Analysis and Rate of Growth, Asymptotic Notation: Big-O Notation, Recursion and Backtracking, Dynamic Programming Strategies, Greedy Algorithms

List of Practicals / Experiments:

List of Practicals

- 1. Introduction to Linux Commands
- 2. Shell Programming
- 3. System Calls (File / Directory/ Process)
- 4. Multi Thread Process using Pthread Library
- 5. SQL Commands (DDL/DML/ TCL)

- 6. PL/SQL Triggers, Procedures, Functions
- 7. Subnetting
- 8. Arrays and Linked List
- 9. Stacks and Queues
- 10. Trees and Binary Search Trees

Text Books:

1. IT INTERVIEW QUESTIONS by NARASIMHA KARUMANCHI, CAREERMONK PUBLICATIONS

References:

1. MCQS IN COMPUTER SCIENCE by TIMOTHY WILLIAMS, MC GRAW HILL

2. SQL IN 10 MINUTES, SAMS TEACH YOURSELF by BEN FORTA, SAMS PUBLISHING

3. CRACKING THE IT INTERVIEW by M BALASUBRAMANIAM, K. R BAALAJI, KIRAN. G RANGANATH, NANDAWAT RAVINDRA K, M SELVAGURU, T COMERICA SUBASH, RAGHAVAN S VENKAT, S ANBAZHAGAN VIKRAM, MC GRAW HILL

4. CRACKING THE CODING INTERVIEW by MCDOWELL GAYLE LAAKMANN, CAREERCUP

CSE332:INDUSTRY ETHICS AND LEGAL ISSUES

L:2 T:0 P:0 Credits:2

Course Outcomes: Through this course students should be able to

CO1 :: apply ethical principles and integrity in professional, business, and IT practices

CO2 :: examine intellectual property concepts and their importance in innovation protection

CO3 :: identify startup schemes, funding sources, and legal support mechanisms.

CO4 :: apply startup planning techniques to develop an IT business plan.

CO5 :: analyze trends, technologies, and diversity in the IT and ITES industry.

CO6 :: analyze ethical, legal, and security issues in information security.

Unit I

Ethics : Definition of ethics, Importance of integrity, Ethics in business world, improving corporate ethics, Creating an ethical work environment, Ethical decision making, Ethics in information technology, Ethical behavior of IT professional, Common ethical issues for IT users, Supporting the ethical practices of IT users, Women Empowerment, Ethical aspects of engineering practices

Unit II

Intellectual Properties(IP'S) : Concept of Intellectual Property, Copyright and Trademark, Different kinds of marks (brand names, logos, signatures, symbols, well known marks, certification marks and service marks, Patents, Importance of Patent Information in Business Development, Identifying social and engineering issues, conducting analysis and registering claims for intellectual property protection

Unit III

Government Funding and Startup schemes : what are Startups, Startup India benefits, Resources, bank loan for start up business, Start-up India, 10000 startups -A NASSCOM Initiative, Export promotion schemes: Software Technology Parks (STPs), Special Economic Zones (SEZ) Scheme, Laws for Startups, SBIR,STTR,NSF Grants

Unit IV

Startup in IT : Planning of startup business in IT sector-Executive summary, General company Description, Products and services, Marketing plan, Operational plan, Management and organization, Personal Financial Statement, Startup Expenses and Capitalization, Financial Plan, Appendices, Refining the Plan, Examples of Successful Start-ups

Unit V

Companies : Introduction to IT and ITES industry (Product based, Services based), Introduction to NASSCOM, STPI, Overview on latest IT projects with global impact, Case study of an IT industry (Product based and Services based), Recent technology advancement, Current affairs related with the IT industry, Diversity in the Workforce

Unit VI

Ethical and Professional issues in Information Security : Law and Ethics in Information Security, Organizational Liability and the Need for Counsel, Policy Versus Law, Cyber Crime, Cyber-crime on the rise, Need for cyber law in India, Ethical dilemmas in project management, Principles of arbitration and alternative dispute resolution, Hands on tools for cyber security, case studies on cyber crimes

References:

1. ETHICS IN INFORMATION TECHNOLOGY by GEORGE W REYNOLDS, CENGAGE LEARNING
2. ETHICS 101 by JOHN .C MAXWELL, Hachette India

MTH265:DISCRETE MATHEMATICS FOR COMPUTING

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: apply the learned concepts to determine equivalence relations, partitions, and partial ordering.

CO2 :: define key concepts such as ordered sets, Hasse diagrams, supremum, infimum, lattices, distributive lattices, and complemented lattices.

CO3 :: explain the representation of Boolean algebra as lattices, the representation theorem, and the significance of the sum-of-products (SOP) form.

CO4 :: explain the concepts of conditional probability, independence (pairwise and mutual), and the binomial distribution in the context of Bernoulli trials.

CO5 :: apply Bayes' theorem, expected values, variance, geometric distribution, and Bienaymé's formula to solve problems involving random variables and probabilistic models.

CO6 :: analyze regular languages, regular expressions, finite state automata, and grammars to classify and solve problems in formal language theory.

Unit I

Relations : introduction, types of relations, reflexive relations, symmetric relations, antisymmetric relations, transitive relations, Closure properties, equivalence relations, partitions, partial ordering relations

Unit II

Ordered Sets and Lattices : introduction, ordered sets, Hasse diagrams of partial ordered sets, consistent enumeration, supremum and infimum, isomorphic (similar) ordered sets, well-ordered sets, lattices and bounded lattices, distributive lattices, complements and complemented lattices

Unit III

Boolean Algebra : introduction, basics definitions, duality, basic theorems, boolean algebras as lattices, representation theorem, sum-of-products form for boolean algebras, minimal boolean expressions, prime implicants

Unit IV

Discrete Probability-I : finite probability, assigning probabilities, probabilities of complements and unions of events, conditional probability, independence, pairwise and mutual independence, Bernoulli trials and the binomial distribution, random variables

Unit V

Discrete Probability -II : Bayes' theorem and generalized Bayes'theorem, expected values, linearity of expectations, the geometric distribution, independent random variables, variance, Bienaymé's formula.

Unit VI

Languages, Automata, Grammars : introduction, alphabet, words, free semigroup, languages and regular languages, regular expressions, finite state automata, grammars and types of grammars

Text Books:

1. DISCRETE MATHEMATICS AND ITS APPLICATIONS(SIE) by KENNETH H ROSEN, MCGRAW HILL EDUCATION

References:

1. DISCRETE MATHEMATICS (SCHAUM'S OUTLINES) (SIE) by SEYMOUR LIPSCHUTZ, MARC LIPSON, VARSHA H. PATIL, MCGRAW HILL EDUCATION

INT396:UNSUPERVISED LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain fundamental concepts of unsupervised learning and real-world problem formulations.

CO2 :: differentiate various unsupervised learning paradigms including hard and soft clustering, partition-based, hierarchical, and density-based approaches.

CO3 :: apply standard unsupervised learning algorithms such as k-Means, Hierarchical Clustering, DBSCAN, GMM, PCA, t-SNE, UMAP, and association rule mining techniques on real-world datasets using Python frameworks.

CO4 :: evaluate clustering, dimensionality reduction, and anomaly detection results using appropriate validation metrics and visualization techniques.

CO5 :: integrate multiple unsupervised learning techniques to construct visual analytics dashboards and exploratory data analysis pipelines.

CO6 :: design a comprehensive unsupervised learning project involving problem identification, data preprocessing, algorithm selection, model evaluation, ethical considerations, and effective presentation of insights.

Unit I

Foundations of Unsupervised Learning : Supervised vs. Unsupervised vs. Semi-Supervised Learning, Problem formulation for unsupervised learning, Real-life use cases: Market segmentation, Customer behavior analysis, Anomaly & fraud detection, Pattern discovery in biological and social data, Distance & similarity metrics:Euclidean, Manhattan, Cosine similarity, when and why distance choice matters

Unit II

Partition-Based Clustering : Clustering fundamentals and assumptions, Hard vs. Soft clustering, k-Means algorithm: Objective function,Initialization strategies (Random, k-Means++), Convergence and limitations, k-Medoids (PAM) vs. k-Means, Data standardization and scaling impact, MiniBatch k-Means for large-scale datasets, Cluster Validation: Inertia (WCSS), Silhouette Coefficient, Davies–Bouldin Index, Elbow method pitfalls

Unit III

Hierarchical & Density-Based Clustering : Hierarchical clustering: Agglomerative vs. Divisive, Linkage methods (single, complete, average, Ward), Dendrogram interpretation, Density-based clustering: DBSCAN fundamentals, e-neighborhood, MinPts, Noise and border points, Comparison: k-Means vs. Hierarchical vs. DBSCAN

Unit IV

Dimensionality Reduction and Representation Learning : Need for Dimensionality Reduction, Linear Dimensionality Reduction Techniques (PCA – Geometric Intuition), Manifold Learning Overview, Non-Linear Dimensionality Reduction Concepts -t-SNE, UMAP – Conceptual, Autoencoder Intuition

Unit V

Association Rule Mining & Anomaly Detection: Association Rule Mining : Support, Confidence, Lift, Conviction, Apriori Algorithm, FP-Growth algorithm, Market Basket Analysis Anomaly Detection: Anomaly vs. novelty detection, Isolation Forest, Local Outlier Factor (LOF), Applications in Cybersecurity and Fraud Detection

Unit VI

Evaluation and Applications of Unsupervised Learning : Internal and External Clustering Validation Metrics, Silhouette Score and Cohesion–Separation Intuition, Stability-Based Evaluation, Interpretability Challenges in Unsupervised Learning, Real-World Case Studies, Ethical Considerations in Pattern Discovery, Case Study : Visualizing handwritten digits (MNIST) or customer segmentation data

Project Component : Create a visualization dashboard comparing PCA vs. t-SNE results

List of Practicals / Experiments:

List of Practicals

- •Implementation and comparison of Euclidean, Manhattan, and Cosine distance metrics on a real-world dataset.

- •Comparative analysis of k-Means and k-Medoids (PAM) clustering on noisy and outlier-prone datasets.
- •Implementation of k-Means clustering from scratch and using scikit-learn.
- •Study of the impact of feature scaling and standardization on k-Means clustering results.
- •Determination of the optimal number of clusters using Elbow Method, Silhouette Score, and Davies–Bouldin Index.
- •Implementation of Hierarchical Clustering using different linkage methods and interpretation of dendrograms.
- •Implementation of DBSCAN and analysis of e (epsilon) and MinPts parameters.
- •Comparative study of k-Means, Hierarchical Clustering, and DBSCAN on the same dataset.
- •Implementation of Principal Component Analysis (PCA) and analysis of variance retention.
- •Visualization of high-dimensional data using PCA and interpretation of principal components.
- •Visualization and comparison of dimensionality reduction techniques using PCA, t-SNE, and UMAP.
- •Implementation of Apriori and FP-Growth algorithms for Market Basket Analysis and interpretation of association rules.
- •Implementation and comparison of Isolation Forest and Local Outlier Factor (LOF) for anomaly detection.
- •End-to-end Unsupervised Learning mini-project including clustering, dimensionality reduction, evaluation, and visualization dashboard.

Text Books:

1. HANDS-ON UNSUPERVISED LEARNING USING PYTHON: HOW TO BUILD APPLIED MACHINE LEARNING SOLUTIONS FROM UNLABELED DATA by ANKUR A. PATEL, O'REILLY

References:

1. KICKSTART UNSUPERVISED MACHINE LEARNING: MASTER UNSUPERVISED MACHINE LEARNING THROUGH PATTERN DISCOVERY, CLUSTERING, AND DIMENSIONALITY REDUCTION TO BUILD INTELLIGENT, REAL-WORLD APPLICATIONS by DR. NIMRITA KOUL, AVA PUBLISHING
2. MODELS AND ALGORITHMS FOR UNSUPERVISED LEARNING: PRACTICAL UNSUPERVISED MACHINE LEARNING by VAIBHAV VERDHAN, Manning Publication

INT396:UNSUPERVISED LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain fundamental concepts of unsupervised learning and real-world problem formulations.

CO2 :: differentiate various unsupervised learning paradigms including hard and soft clustering, partition-based, hierarchical, and density-based approaches.

CO3 :: apply standard unsupervised learning algorithms such as k-Means, Hierarchical Clustering, DBSCAN, GMM, PCA, t-SNE, UMAP, and association rule mining techniques on real-world datasets using Python frameworks.

CO4 :: evaluate clustering, dimensionality reduction, and anomaly detection results using appropriate validation metrics and visualization techniques.

CO5 :: integrate multiple unsupervised learning techniques to construct visual analytics dashboards and exploratory data analysis pipelines.

CO6 :: design a comprehensive unsupervised learning project involving problem identification, data preprocessing, algorithm selection, model evaluation, ethical considerations, and effective presentation of insights.

Unit I

Foundations of Unsupervised Learning : Supervised vs. Unsupervised vs. Semi-Supervised Learning, Problem formulation for unsupervised learning, Real-life use cases: Market segmentation, Customer behavior analysis, Anomaly & fraud detection, Pattern discovery in biological and social data, Distance & similarity metrics:Euclidean, Manhattan, Cosine similarity, when and why distance choice matters

Unit II

Partition-Based Clustering : Clustering fundamentals and assumptions, Hard vs. Soft clustering, k-Means algorithm: Objective function,Initialization strategies (Random, k-Means++), Convergence and limitations, k-Medoids (PAM) vs. k-Means, Data standardization and scaling impact, MiniBatch k-Means for large-scale datasets, Cluster Validation: Inertia (WCSS), Silhouette Coefficient, Davies–Bouldin Index, Elbow method pitfalls

Unit III

Hierarchical & Density-Based Clustering : Hierarchical clustering: Agglomerative vs. Divisive, Linkage methods (single, complete, average, Ward), Dendrogram interpretation, Density-based clustering: DBSCAN fundamentals, e-neighborhood, MinPts, Noise and border points, Comparison: k-Means vs. Hierarchical vs. DBSCAN

Unit IV

Dimensionality Reduction and Representation Learning : Need for Dimensionality Reduction, Linear Dimensionality Reduction Techniques (PCA – Geometric Intuition), Manifold Learning Overview, Non-Linear Dimensionality Reduction Concepts -t-SNE, UMAP – Conceptual, Autoencoder Intuition

Unit V

Association Rule Mining & Anomaly Detection: Association Rule Mining : Support, Confidence, Lift, Conviction, Apriori Algorithm, FP-Growth algorithm, Market Basket Analysis Anomaly Detection: Anomaly vs. novelty detection, Isolation Forest, Local Outlier Factor (LOF), Applications in Cybersecurity and Fraud Detection

Unit VI

Evaluation and Applications of Unsupervised Learning : Internal and External Clustering Validation Metrics, Silhouette Score and Cohesion–Separation Intuition, Stability-Based Evaluation, Interpretability Challenges in Unsupervised Learning, Real-World Case Studies, Ethical Considerations in Pattern Discovery, Case Study : Visualizing handwritten digits (MNIST) or customer segmentation data

Project Component : Create a visualization dashboard comparing PCA vs. t-SNE results

List of Practicals / Experiments:

List of Practicals

- •Implementation and comparison of Euclidean, Manhattan, and Cosine distance metrics on a real-world dataset.

- •Comparative analysis of k-Means and k-Medoids (PAM) clustering on noisy and outlier-prone datasets.
- •Implementation of k-Means clustering from scratch and using scikit-learn.
- •Study of the impact of feature scaling and standardization on k-Means clustering results.
- •Determination of the optimal number of clusters using Elbow Method, Silhouette Score, and Davies–Bouldin Index.
- •Implementation of Hierarchical Clustering using different linkage methods and interpretation of dendrograms.
- •Implementation of DBSCAN and analysis of e (epsilon) and MinPts parameters.
- •Comparative study of k-Means, Hierarchical Clustering, and DBSCAN on the same dataset.
- •Implementation of Principal Component Analysis (PCA) and analysis of variance retention.
- •Visualization of high-dimensional data using PCA and interpretation of principal components.
- •Visualization and comparison of dimensionality reduction techniques using PCA, t-SNE, and UMAP.
- •Implementation of Apriori and FP-Growth algorithms for Market Basket Analysis and interpretation of association rules.
- •Implementation and comparison of Isolation Forest and Local Outlier Factor (LOF) for anomaly detection.
- •End-to-end Unsupervised Learning mini-project including clustering, dimensionality reduction, evaluation, and visualization dashboard.

Text Books:

1. HANDS-ON UNSUPERVISED LEARNING USING PYTHON: HOW TO BUILD APPLIED MACHINE LEARNING SOLUTIONS FROM UNLABELED DATA by ANKUR A. PATEL, O'REILLY

References:

1. KICKSTART UNSUPERVISED MACHINE LEARNING: MASTER UNSUPERVISED MACHINE LEARNING THROUGH PATTERN DISCOVERY, CLUSTERING, AND DIMENSIONALITY REDUCTION TO BUILD INTELLIGENT, REAL-WORLD APPLICATIONS by DR. NIMRITA KOUL, AVA PUBLISHING
2. MODELS AND ALGORITHMS FOR UNSUPERVISED LEARNING: PRACTICAL UNSUPERVISED MACHINE LEARNING by VAIBHAV VERDHAN, Manning Publication

:

L: T: P: :

Course Outcomes: Through this course students should be able to

CO1 ::